

# Kitonga Farm Villas & Agro-Tourism Management System

Comprehensive engineering documentation, system architecture breakdown, database schema, REST API mapping, point-of-sale operations, and standard operating procedures (SOP).

| VERSION            | STACK              | DATABASE            | LOCATION              |
|--------------------|--------------------|---------------------|-----------------------|
| 2.0.0 (Production) | Laravel 12 + Vue 3 | MySQL 8 (41 Tables) | Handeni / Kilimanjaro |

## 1 Executive Summary & Vision

**Kitonga Farm Villas** is a flagship luxury eco-hospitality resort and commercial agro-enterprise. The platform fuses high-end eco-villa accommodation booking with real-time farm operations, point-of-sale commerce, agro-tour reservations, and financial accounting into a unified web ecosystem.



### Guest Experience & Bookings

Explore villa sanctuaries, reserve tours, and order farm harvest with zero friction



### Counter POS & Real-Time Stock

and instant receipting.

Fast cashier checkout for milk, mtindi, yogurt, honey, and eggs with automated warehouse stock decrements.

## 2 Architectural Framework & Stack

- **Backend Core:** Laravel 12.x running on PHP 8.2+ with Strict Types and Eloquent ORM.
- **Frontend Single Page Application (SPA):** Vue 3 (Composition API) powered by Inertia.js v2 protocol.
- **Asset Compilation:** Vite 7.x with dynamic code-splitting and asset minification.
- **Styling:** Tailwind CSS with custom editorial brand tokens (`#14231C`, `#FAF8F5`, `#C98A3E`).
- **Database Engine:** MySQL 8.0+ (`kitonga\_villas\_db`) featuring 41 relational tables with B-Tree indexing.
- **Role-Based Access Control (RBAC):** Spatie Laravel-Permission module.

## 3 Core Modules & Functionality

### 1. Villa Booking & Accommodation Engine

Dynamic availability matrix, double-booking prevention through atomic concurrency locks, automated invoice generation, and seasonal rate calculation.

### 2. POS Counter Terminal

Live touch interface, SKU/barcode fast searching, multi-gateway tender (Cash, M-Pesa, Card, Room Charge), change calculation, and 80mm/58mm thermal receipt printing.

### 3. Farm Produce & Dairy Inventory

Catalog tracking for all 12 authentic farm products (Mtindi 5L/3L, Fresh Milk 5L/3L, Yogurt 1L/0.5L, Eggs, Forest Honey, Mangoes, Papaws, Pineapples, Vegetables) with low-stock warnings.

#### 4. Financial Ledger & Expense Tracking

Automated gross revenue tracking, operational expenditure logging by category, and instant net operating profit calculation.

### 4 Authentic Farm Produce & Price List

| SKU               | PRODUCT NAME                             | CATEGORY         | UNIT           | PRICE (TZS) |
|-------------------|------------------------------------------|------------------|----------------|-------------|
| KFV-MTINDI-5L     | Kitonga Cultured Sour Milk / Mtindi (5L) | Dairy & Eggs     | 5 Liters       | 17,000      |
| KFV-MTINDI-3L     | Kitonga Cultured Sour Milk / Mtindi (3L) | Dairy & Eggs     | 3 Liters       | 13,000      |
| KFV-FRESH-5L      | Fresh Whole Farm Milk (5L)               | Dairy & Eggs     | 5 Liters       | 13,000      |
| KFV-FRESH-3L      | Fresh Whole Farm Milk (3L)               | Dairy & Eggs     | 3 Liters       | 9,000       |
| KFV-YOGURT-1L     | Kitonga Farm Artisanal Yogurt (1L)       | Dairy & Eggs     | 1 Liter        | 6,000       |
| KFV-YOGURT-05L    | Kitonga Farm Artisanal Yogurt (0.5L)     | Dairy & Eggs     | 0.5 L          | 3,000       |
| KFV-MAYAI-KISASA  | Farm Fresh Organic Eggs (Tray of 30)     | Dairy & Eggs     | Tray (30 Eggs) | 8,000       |
| KFV-ASALI-1KG     | Raw Wild Forest Honey (1kg)              | Honey            | 1 kg           | 10,000      |
| KFV-MANGO-1KG     | Sweet Kitonga Highland Mangoes (1kg)     | Fruits           | 1 kg           | 3,000       |
| KFV-PAPAW-1PC     | Tree-Ripened Sweet Papaws                | Fruits           | Piece / kg     | 4,000       |
| KFV-PINEAPPLE-1PC | Sun-Drenched Estate Pineapples           | Fruits           | Piece          | 4,500       |
| KFV-VEG-BUNDLE    | Spring-Fed Fresh Vegetables              | Fresh Vegetables | Bundle         | 6,000       |

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## Database Schema Architecture

The system runs on the relational database ``kitonga_villas_db`` with 41 tables:

- `accommodation_types` (Villa Categories)

- `accommodation_units` (Individual Villa Rooms)

- `bookings` (Guest Reservations & Dates)

- `payments` (Financial Tender Records)

- `products` (Dairy & Farm Stock Items)

- `sales` (POS Transaction Headers)

- `sale_items` (Line Items for Counter Sales)

- `farm_tours` (Experiences & Tours)

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## Production Deployment & Operations

### # Production Optimization Commands:

```
composer install --no-dev --optimize-autoloader
npm run build
php artisan config:cache
php artisan route:cache
php artisan view:cache
```